The force exerted by the block m_1 on m_2 due to its motion is called force of contact ($\vec{f}_{21}$). According to Newton's third law, the block m_2 will exert an equivalent opposite reaction force ($\vec{f}_{12}$) on block m_1.

Figure 3.14 (b) shows the free body diagram of block m_1.

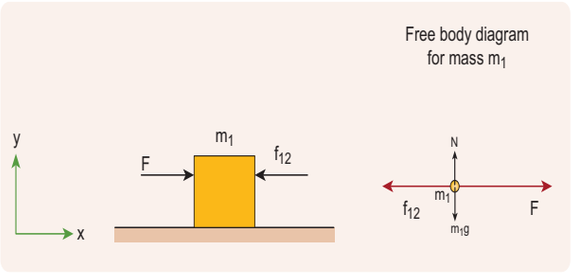


Figure 3.14 (b) Free body diagram of block of mass m_1

$$\therefore F\hat{i} - f_{12}\hat{i} = m_1 a\hat{i}$$

By comparing the components on both sides of the above equation, we get

$$F - f_{12} = m_1 a$$

$$f_{12} = F - m_1 a \qquad (3.6)$$

Substituting the value of acceleration from equation (3.5) in (3.6) we get

$$f_{12} = F - m_1\left(\frac{F}{m_1 + m_2}\right)$$

$$f_{12} = F\left[1 - \frac{m_1}{m_1 + m_2}\right]$$

$$f_{12} = \frac{Fm_2}{m_1 + m_2} \qquad (3.7)$$

Equation (3.7) shows that the magnitude of contact force depends on mass m_2 which provides the reaction force. Note that this force is acting along the negative x direction.

In vector notation, the reaction force on mass m_1 is given by $\vec{f}_{12} = -\frac{Fm_2}{m_1 + m_2}\hat{i}$

For mass m_2 there is only one force acting on it in the x direction and it is denoted by $\vec{f}_{21}$. This force is exerted by mass m_1. The free body diagram for mass m_2 is shown in Figure 3.14 (c).

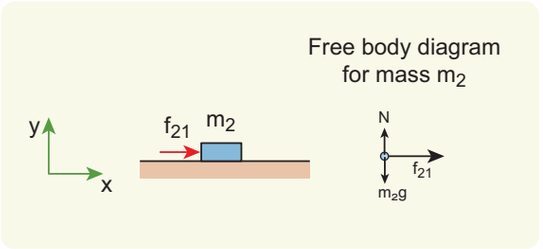


Figure 3.14 (c) Free body diagram of block of mass m_2

Applying Newton's second law for mass m_2 $\quad f_{21}\hat{i} = m_2 a\hat{i}$

By comparing the components on both sides of the above equation

$$f_{21} = m_2 a \qquad (3.8)$$

Substituting for acceleration from equation (3.5) in equation (3.8), we get $f_{21} = \frac{Fm_2}{m_1 + m_2}$

In this case the magnitude of the contact force is

$f_{21} = \frac{Fm_2}{m_1 + m_2}$ The direction of this force is along the positive x direction.

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